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Recognition of promising technologies considering inventor and assignee's historic performance: A machine learning approach

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ABSTRACT

Recognition of promising technologies is important for enterprises that are eager to occupy the center position in future markets, considering it can provide valuable R&D (research and development) intelligence and industrial reform signal. Accordingly, many approaches have been introduced in the literature to identify promising technologies; however, most previous studies have relied heavily on technologies' bibliometric and text features. The promisingness of a technology is often determined by various other factors, including the inventor and assignee's historic performance (IAHP) features. To overcome the limitations of previous approaches, we propose a machine learning approach that integrates the bibliometric, text, and IAHP features to accomplish promising technology recognition (named MLIFPR) in this paper. The proposed MLIFPR approach was applied to five fields, including the electrical communication, ship, road construction, electric power, and electric vehicle fields, and its usability was verified. Experiments showed that the approach considering bibliometric, text, and IAHP features achieved an average promising technology recognition precision of 95.9 %, which outperformed existing studies. The recognition performance was significantly improved by >3.3 % after the addition of IAHP features. The proposed MLIFPR approach of this study is a novel data mining tool to assist enterprises in developing and following unhatched disruptive technologies to occupy a leading position in the future market. Besides, the introduced IAHP features as new low-dimensional signals representing the technologies' promisingness can provide a reference for other technological innovation works.

1. Introduction

In order to cope with current rapid technological change and fierce market competition, enterprises are increasingly eager to capture relevant signals of promising technologies. This is essential for developing effective research and development (R&D) strategies and avoiding obsolescence in the market. As promising technologies, or next-generation disruptive technologies, have the potential to drive industrial revolutions by designing innovative and competitive products that generate substantial commercial profits and reshape market dynamics (Lee and Sohn, 2021; Haghighian Roudsari et al., 2022). For instance, Huawei's "Polar Codes" (Arikan, 2009) technology has significantly advanced the communication market from 4G to 5G, while MIT's "Immersion Liquid" (Switkes and Rothschild, 2001) technology has transformed the photolithography industry from dry to immersion lithography.

Previous studies have proposed various approaches for recognizing promising technologies, which can be categorized into three main aspects: promising technology recognition (Trappey et al., 2021; Choi et al., 2020), analysis of technological evolution trends (Liu et al., 2024), and identifying technology convergence (Kim and Sohn, 2020). The first aspect involves extracting multiple bibliometric and text indicators from patents as predictors, employing machine learning models to assess the promisingness of the technologies contained within those patents. The second aspect focuses on collecting patent citation relationships to uncover rising technologies hidden within historical technological evolution. Finally, the last aspect utilizes complex network analysis or machine learning methods to forecast potential technology convergence, thereby supporting R&D activities related to the recognition of promising technologies.

Despite their value, most of existing studies tend to emphasize bibliometric and text information from patents (Mukherjee et al., 2017)

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while neglecting other factors that could enhance the performance of promising technology recognition. Notably, the inventors and assignees' historical performance (IAHP) features, which map R&D capacity, have been shown to effectively reflect the promisingness of a technology (Zhu et al., 2019; Jaemin et al., 2021). Therefore, it is essential to incorporate IAHP features as predictive factors in the promising technology recognition task. However, the extraction of IAHP features from extensive unstructured patent metadata and their integration into the promising technology recognition framework presents a significant challenge for technological innovation works.

In light of these considerations, this paper proposes a machine learning approach—referred to as MLIFPR—that integrates bibliometric, text, and IAHP features from patents for the recognition of promising technologies. We validate this approach using patent data from the fields of "Electrical Communication," "Ship," "Construction Road," "Electric Power," and "Electric Vehicle," sourced from the China National Intellectual Property Administration (CNIPA), the United States Patent and Trademark Office (USPTO), and the European Patent Office (EPO). Our experimental results indicate that incorporating IAHP features significantly enhances the performance of promising technology recognition. Moreover, the fusion of bibliometric, text, and IAHP features further improves the reliability of the recognition task.

In particular, we make the following contributions:

1) We propose an MLIFPR framework that fuses multidimensional features (bibliometric, text, and IAHP) for the recognition of promising technologies, assisting governments and enterprises in discovering, monitoring, and developing emerging disruptive technologies, thereby avoiding obsolescence in the next technological revolution.
2) This study offers a quantitative model to capture the R&D capabilities of inventors and assignees (IAHP information), which can serve as a reference for other technological innovation works, including technology fusion prediction.

The remainder of this study is organized into four parts. Section 2 contains a *literature review* that provides the background for this study, and promising technology recognition approaches have been proposed in previous studies. In Section 3, the framework of the *promising technology recognition approach* is described, and the general and detailed processes involved, including dataset collection, feature extraction, and experiment, are discussed. In Section 4, a *discussion* that includes some implications of the MLIFPR approach where it is likely to be helpful for R&D workers is listed. Finally, a *conclusion*, in which the contributions and limitations of this study are summarized, is presented in Section 5.

2. Literature review

2.1. Promising technology recognition

Existing studies on promising technology recognition primarily utilize bibliometric information from patents as predictive factors (Kyebambe et al., 2017; Zhou et al., 2020). Many patent bibliometric indicators that do not require extensive data accumulation are extracted as the input features of technologies. Machine learning models are then trained to discern the differences between recognized promising technologies and non-promising technologies. For instance, existing studies have utilized various bibliometric indicators, such as the technological breadth of patents (Trappey et al., 2021) and the number of patents within specific technical fields (Choi et al., 2020), to forecast promising technologies. Additionally, other studies highlight the importance of incorporating text (Arts et al., 2021) and assignee (Lee et al., 2018) information into the promising technology recognition process. These studies employ multiple indicators related to text and assignees, such as text embedding vectors (Lee et al., 2020), the number of patents held by assignees (Kwon and Geum, 2020), and forward citations of assignees' patents (Choi et al., 2021).

2.2. Analysis of technological evolution trend

In addition to promising technology recognition, other methods for identifying promising technologies involve analyzing technological evolution trends. Given that patent citation relationships reflect technological flows and developments, many existing studies utilize patent citation network analysis to capture technological evolution trends. For example, various patent analysis methods have been employed to investigate patent citation networks and explore technological evolution, including main path analysis (Watanabe and Takagi, 2021) and complex network analysis (Zamani et al., 2022). Furthermore, recent studies have identified that patent text information plays a crucial role in discovering technological evolution trends (Xi et al., 2023; Oh et al., 2023). Additionally, some researchers have expanded their data sources from patents to scientific papers to enhance the accuracy of technological evolution trend forecasts (Liu et al., 2024; Chen et al., 2024).

2.3. Identifying technology convergence

The identification of technology convergence is another area of research closely related to promising technology recognition. Patent classification information, which indicates the technological fields in which patents reside, is essential for determining whether technology convergence has occurred. Technology convergence is evidenced when two or more patent classification numbers appear in the same patent. Consequently, existing studies frequently utilize patent classification number co-occurrence information to survey technology convergence (Caviggioli, 2016; Jeong et al., 2015). To address the growing demand for predicting technology convergence, many studies employ complex network analysis and machine learning approaches to anticipate future links within the patent classification co-occurrence network. For example, existing studies utilize multiple network indicators—such as the common neighbor indicator, resource allocation indicator, and local path indicator—to predict future links in the patent classification number co-occurrence network, aiming to uncover insights into technology convergence (Ma et al., 2022; Lee et al., 2022; Duan and Guan, 2021). Moreover, existing research has demonstrated that incorporating text information significantly enhances the performance of technology convergence forecasts (Kim and Sohn, 2020; Xu et al., 2022; Ebadi et al., 2022).

In summary, although existing methods can recognize promising technologies, their performance needs to be improved. Most of them focus primarily on bibliometric and text features of technologies (Reitzig, 2004; Zhang et al., 2017; Andrew et al., 2015; Arts et al., 2021; Noh et al., 2015) and give limited attention to other factors influencing technological promisingness. The IAHP features, which reflect the R&D capabilities and professional knowledge accumulation of inventors and assignees, can enhance the performance of promising technology recognition tasks. The greater the R&D capabilities of an inventor or assignee, the higher the potential for a technology to be deemed promising (Chen and Chang, 2010; Du et al., 2021; Choi and Lee, 2020; Liu et al., 2008). Thus, collecting and utilizing IAHP features is vital for recognizing promising technologies. Consequently, we extract multidimensional features—encompassing bibliometric, text, and IAHP information—to facilitate the recognition of promising technologies.

3. Promising technology recognition approach

Fig. 1 shows the framework of the proposed MLIFPR approach, which employs a machine learning approach based on multidimensional feature fusion to identify promising technologies. The overall research process used to implement the framework consists of three steps (see Fig. 1). In the first step (Fig. 1a), we download patents' bibliometric

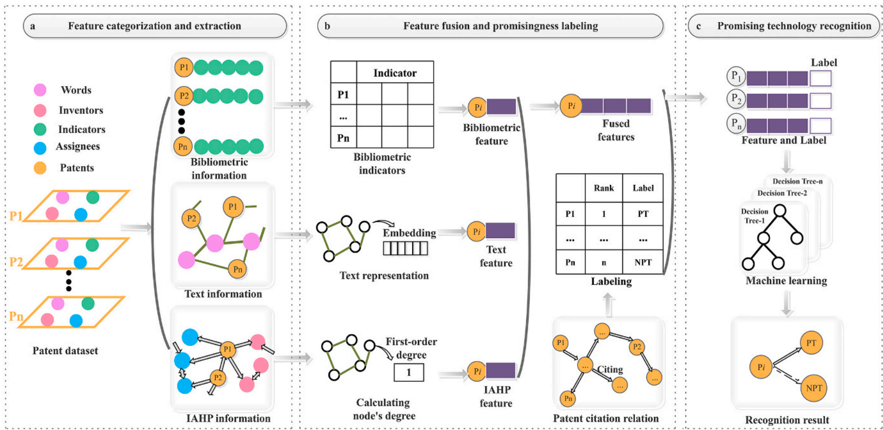

Fig. 1. The framework of the MLIFPR approach.

indicators, text, inventors, and assignees from the [1]PatSnap website and transform them into structured bibliometric, text, and IAHP feature vectors of technologies, which are used to train our machine learning models. In the next step (Fig. 1b), the obtained technologies' features are fused, and the patent citation relation is employed to label the technologies' promisingness. The final step (Fig. 1c) involves inputting the features and labels of technologies to train the machine learning model and using the trained model to recognize promising technologies. Considering the large number of acronyms in this paper, we have provided an acronym table in Appendix 1 (see Table A1 in Appendix 1) for fluid reading.

3.1. Collection and preprocessing of dataset

Given that technologies are published in the form of patents, we need to collect patents for extracting their features in the dataset collection stage. To test the performance of the MLIFPR approach, we collect four fields' patents examined by the CNIPA from the PatSnap website, including the electrical communication, electric power, construction road, and ship fields, considering they are the pillar industries in China. Besides, we also collect patents in the rapidly developed electric vehicle field from the USPTO and EPO database of PatSnap website to test the applicability of our proposed MLIFPR approach. These fields' descriptions, retrieval types, and the number of collected patents are shown in Table 1.

As the 5-years forward citations of patents used for labeling patents' promisingness need time to accumulate, we select patents published before 2016. In addition, the number of patents in our selected five fields before 2000 is small, thus, we only choose patents published after 2000. From the PatSnap website, we can retrieve nearly all published patents in each field, along with their publication numbers and essential information, including various structured bibliometric indicators, text data, patent inventors, and patent assignees. This information is then downloaded for use in our promising technology recognition tasks.

In the process of dataset collection, we find that the published patent does not provide a unique identifier for the inventor, hence the inventor's name disambiguation is necessary. In this study, referring to existing studies (Sinatra et al., 2016; Xiaomei et al., 2019), inventors are considered to be the same one if the following two conditions are met at the same time: (1) the inventors' names are the same, and (2) the inventors have the entrustment relationship with the same assignee. Our inventor's name disambiguation method can also disambiguate Chinese and English languages, only need to extra translate Chinese names to English when judging whether Chinese names and English names are the same.

Table 1
The basic information of the dataset.

Field	Description	Retrieval type	Patent count
Electrical communication (EC)	This field contains patents related to electrical communication technology, such as broadcast communication, transmission of digital information, and wireless communication.	APD:[20000101 TO 20151231] AND IPC:(H04)	285,257
Ship	This field contains patents related to ships or other waterborne vessels and their related equipment, such as waterborne vessels for shipping, offensive or defensive arrangements on vessels, and auxiliaries on vessels.	APD:[20000101 TO 20151231] AND IPC:(B63)	8202
Road construction (RC)	This field contains patents related to the construction of roads, such as railways, bridges, and street cleaning.	APD:[20000101 TO 20151231] AND IPC:(E01)	18,104
Electric power (EP)	This field contains patents related to electric power, such as the generation of electric power and transmission of electric power.	APD:[20000101 TO 20151231] AND IPC:(H02)	99,983
Electric vehicle_uspto (EV_uspto)	This field contains patents related to electric vehicle in the USPTO database.	TAC:(electric vehicle) AND PBD:[20000101 TO 20151231]	37,314
Electric vehicle_epo (EV_epo)	This field contains patents related to electric vehicle in the EPO database.	TAC:(electric vehicle) AND PBD:[20000101 TO 20151231]	20,025

The process of inventors' name disambiguation is shown in Fig. 2. Specifically, we extract the inventor and assignee information of each patent in our selected dataset to generate a data frame, and then construct a null list that contains the disambiguated name of inventors. Next, for each inventor, we check whether his name is on our list. If not, we carry out the disambiguation operation. First, we add a number after the inventor's name and query the data frame. If we encounter an inventor with the same name, we check whether the assignee is consistent. If the assignee is consistent, we add the same number after the inventor's name. If the assignee is inconsistent, we skip the inventor and continue

[1] https://www.zhihuiya.com/

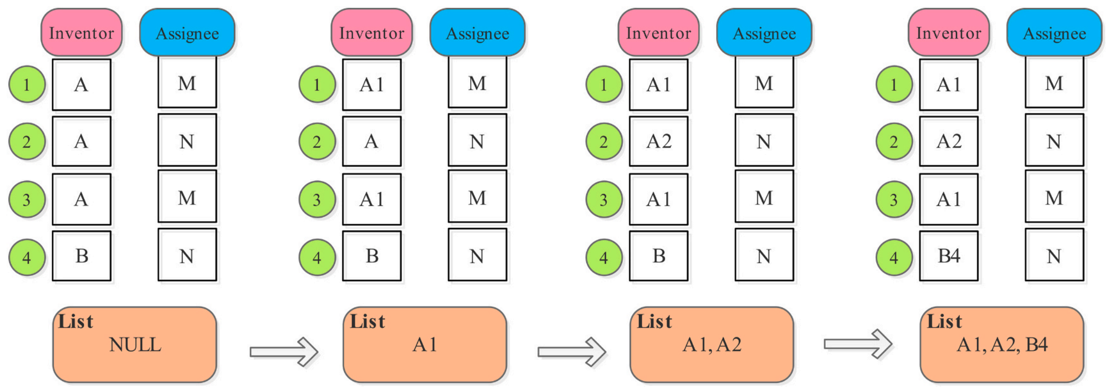

Fig. 2. The process of inventors' name disambiguation.

the query until all inventors are checked. Finally, we repeat the above process to complete the disambiguation of all inventor's names.

3.2. Feature extraction

In the feature extraction stage, we divide selected technologies' features into three categories including bibliometric, text, and IAHP features based on their information source. For each category of technologies' features, we describe their extraction process and the reasons for selecting them. Specifically, we directly extract bibliometric features from the PatSnap website, employ the natural language process model to preprocess and vectorize text features, and construct the network model to obtain IAHP features.

3.2.1. Extracting bibliometric features

There are numerous bibliometric indicators in patents that can represent technologies' promisingness, according to existing literature (see Table 2). Specifically, a total of 10 indicators are employed to cover the bibliometric features of technologies, which can be categorized into three aspects. (1) *Basic information.* This subcategory contains a total of four indicators, including NWFC, NC, NI, and NA. Referring to existing studies listed in Table 2, we argue that promising technologies usually possess more NWFC and NC. As abundant patent claims can assist assignees gain more competitive edges in the market. Besides, we think that NI and NA have a positive correlation with the promisingness of technologies. Because they represent the potential that a technology becomes promising technology. (2) *Protection scope information.* The protection range of technology refers to filing the technology in more countries to acquire more commercial profit and market competitiveness. In this subcategory, the proposed approach herein uses NIPC, NSFP, NEFP, and NPP to represent technologies' promisingness. Referring to existing studies listed in Table 2, we argue that promising technologies usually possess more NIPC, NSFP, NEFP, and NPP. Enhancing these indicators can assist assignees in acquiring more commercial profit from different countries and regions. (3) *Backward citation information.* This subcategory comprises a total of two indicators, namely, NBC and NCP. Referring to existing studies listed in Table 2, we argue that NBC and NCP have a positive correlation with the promisingness of technologies. Because more technological knowledge accumulation can improve technologies' potential to become promising technologies.

3.2.2. Extracting IAHP features

The IAHP features of technologies are indispensable in the feature extraction process considering they can reflect the promisingness of technologies (references are listed in Table 2). Therefore, we collect the IAHP features of technologies, abstract them into indicators, and offer their extraction method.

Specifically, in the first step, we construct a heterogeneous network by collecting the inventor and assignee information of each technology. In the heterogeneous network, technologies, inventors, and assignees are represented as nodes. Citation relationships, invention relationships, application relationships, cooperation relationships, transfer relationships, and entrustment relationships are represented as edges. In the second step, based on this heterogeneous network, we calculate the number of inventors' relationships and assignees' relationships to obtain their historical performance, namely the value of each IAHP indicator. For each inventor, we calculate their number of invention relationships and cooperation relationships to obtain the value of NPII and NCI indicators, respectively. Then, we use Formula (1), Formula (2), and Formula (3) to obtain the value of their SCPI, MCPI, and HCPI indicators. For each assignee, we calculate their number of application relationships, transfer relationships, and entrustment relationships to obtain the value of NPFA, NTPA, and NIEA indicators. Then, we use Formula (1), Formula (2), and Formula (3) to obtain the value of their SCPA, MCPA, and HCPA indicators.

$$scp = \sum_{i=1}^{n} c_i \tag{1}$$

$$mcp = \sum_{i=1}^{n} c_i \Big/ n \tag{2}$$

$$hcp = max(\{c_1, c_2, \dots, c_n\}) \tag{3}$$

where n is the number of inventors' invention relationships or assignees' application relationships, c_i is the number of citation relationships of inventors' or assignees' technologies.

Fig. 3 provides a specific diagram of a constructed heterogeneous network. In Fig. 3, it can be observed that inventors, assignees, and technologies are represented as different color nodes, and relationships between them are represented as different color edges. For inventor I_1, we observe he possesses two invention relationships (pink edges) and one cooperation relationship (purple edge), thus the value of his NPII and NCI is 2 and 1. In addition, the citation relationships (cyan edges) of technology P_1 and technology P_3 are one and two, thus the value of his SCPI, MCPI, and HCPI is 3, 1.5, and 2, respectively. For assignee A_1, it has been identified that he possesses three application relationships (blue edges), one transfer relationship (orange edge), and one entrustment relationship (green edge), thus the value of his NPFA, NTPA, and NIEA is 3, 1, and 1, respectively. In addition, the citation relationships (cyan edges) of technology P_1, P_4, and technology P_5 are one, zero, and zero, thus the value of his SCPA, MCPA, and HCPA is 1, 0.3 and 1, respectively.

In the third step, for each technology, only IAHP indicators of the first inventor and the first assignee are calculated as its features. Based on the constructed heterogeneous network, a total of 11 indicators are employed to cover the IAHP features of technologies, which can be categorized into two aspects. (1) *Inventor's historic performance*

Table 2
Indicators used for representing the promisingness of a technology.

Feature	Indicator	Description	Reference
bibliometric	NWFC (The number of words in the first claim) NC (The number of claims) NA (The number of assignees) NI (The number of inventors)	Basic information	Breitzman and Thomas, 2015; Fischer and Leidinger, 2014; Milanez et al., 2017
	NIPC (The number of International Patent Classifications) NSFP (The number of simple family patents) NEFP (The number of extended family patents) NPP (The number of priority patents)	Protection scope information	Lanjouw and Schankerman, 2004; Squicciarini et al., 2013; Harhoff et al., 2003
	NBC (The number of backward citations) NCP (The number of citing papers)	Backward citation information	Callaert et al., 2006; Poege et al., 2019
IAHP	NPII (The number of patents invented by the inventor) SCPI (The sum of forward citations of patents of the inventor) MCPI (The mean of forward citations of patents of the inventor) HCPI (The highest forward citations of patents of the inventor) NCI (The number of coinventors)	Inventor's historical performance information	Grigoriou and Rothaermel, 2014; Bass and Kurgan, 2010; Ajay et al., 2020; Ferrara et al., 2017
	NPFA (The number of patents filed by the assignee) SCPA (The sum of forward citations of patents of the assignee) MCPA (The mean of forward citations of patents of the assignee) HCPA (The highest of forward citations of patents of the assignee) NTPA (The number of transferred patents of the assignee) NIEA (The number of inventors entrusted to the assignees)	Assignee's historic performance information	Bessen, 2008; Gambardella and Harhoff, 2008; Funk and Owen-Smith, 2017
text	$Topic_0$ $Topic_k$	Semantic information	Lee et al., 2020; Han and Sohn, 2015; Gustaf et al., 2020; Arts et al., 2021.

information. This subcategory comprises a total of five indicators, that is, NPII, SCPI, MCPI, HCPI, and NCI. Referring to existing studies listed in Table 2, we argue that NPII, SCPI, MCPI, HCPI, and NCI have a positive correlation with the promisingness of technologies. As they represent the R&D capability of inventors from multiple perspectives, including workload, peer recognition, and social interaction. (2) *Assignee's historic performance information*. This subcategory comprises six indicators that include NPFA, SCPA, MCPA, HCPA, NTPA, and NIEA. Referring to existing studies listed in Table 2, we argue that promising technologies usually possess more NPII, SCPI, MCPI, HCPI, and NCI. Given that they can represent the R&D capability of assignees from various aspects, including workload, peer recognition, human resources, and market recognition.

3.2.3. Extracting text features

The text information is also vital for the promising technology recognition task (see references listed in Table 2). Because the promisingness of technologies often is represented into attractive text for obtaining authorization. Thus, we merge the title, claim, and abstract of the technology into a text, and then process it into the text features. Specifically, our approach employs the latent Dirichlet allocation (LDA) model to determine the optimal topic number of each selected dataset, and then obtain the topic probability of each technology as its text features. This processing method can enhance the explainability of text features. Besides, we also employ the text representation model to obtain the text embedding vector of technologies to complete the promising technology recognition task. Although many text representation models including TextCNN (Zhang and Wallace, 2015), TextGCN (Yao et al., 2018), and Bert (Devlin et al., 2018), can process text data, our approach selects the Doc2Vec (Le and Mikolov, 2014) model because of its simplicity and effectiveness (Lau and Baldwin, 2016; Kim et al., 2018).

Table 2 provides a summary of all collected indicators used for representing technologies' promisingness. Here, it should be noted that our approach only uses indicators that may be obtained immediately after patent publication and do not require time to accumulate data. Hence, our approach is adapted for the recognition of promising technology contained in newly published patents.

3.3. Modeling

Besides collecting the bibliometric, text, and IAHP features of technologies, we need to assign the promising or nonpromising label for each technology before training the MLIFPR approach. Given that the patent application does not point out a technology's promisingness. Although various indicators (e.g., the number of forward citations, the lifetime of the patent) can be used for labeling technologies' promisingness (Choi et al., 2020; Lee and Sohn, 2017; Wu et al., 2019), one of the most frequently employed indicators is the number of forward citations of a technology (Choi et al., 2021; Kwon and Geum, 2020; Li et al., 2018). Because many studies have proven that the number of forward citations is a reliable proxy of technologies' promisingness (Chai et al., 2020; Moser et al., 2017; Arts et al., 2012; Bakker et al., 2016; Atallah and Rodriguez, 2006; Gambardella and Harhoff, 2008).

The labeling process is as follows. First, we calculate the number of 5-year forward citations of each technology (cumulative forward citations during 5 years after each technology's published year). Next, technologies in the top 10 % of the 5-year forward citations rank serve as promising technologies and are assigned the label 1, the remaining technologies serve as nonpromising technologies and are assigned the label 0 (refers to Choi et al., 2021; Kwon and Geum, 2020). Finally, the promising technology recognition is converted into a classification task.

Then we divide the dataset into a training dataset and a test dataset in an 8:2 ratio for training the MLIFPR approach. Given that the proportion (1:9) of PTs (promising technologies) and NPTs (nonpromising technologies) in our dataset is not balanced, most machine learning classification models prefer the majority class in an imbalanced dataset (Lei et al., 2022), we apply SMOTE (synthetic minority over-sampling technique) (Chawla et al., 2002) to generate balanced training sets (Weis and Jacobson, 2021).

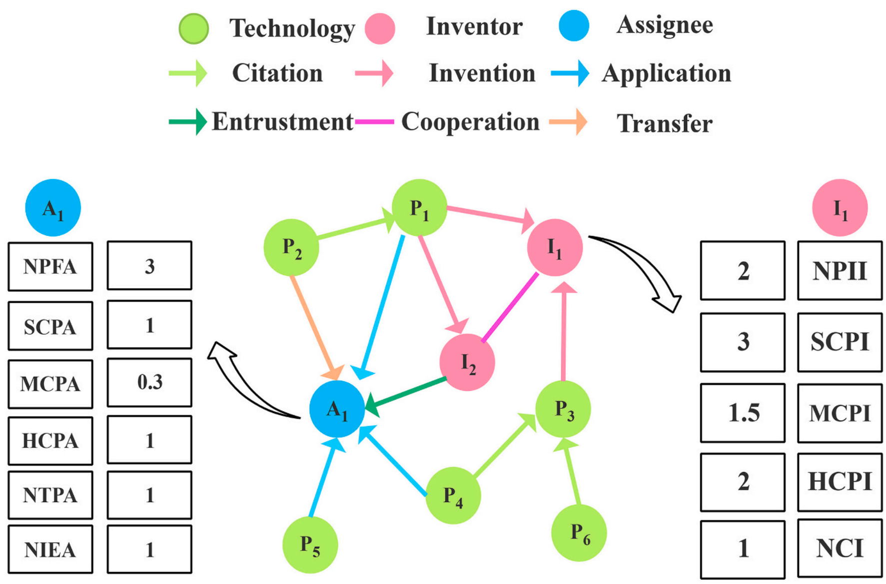

Fig. 3. The diagram of calculating IAHP indicators.

To evaluate the performance of the MLIFPR approach in promising technology recognition tasks, a confusion matrix is used and lists the comparison of the results of the actual and predicted classes, as shown in Table A6 in Appendix 3. Based on the confusion matrix, three performance measuring metrics are calculated, including precision (Formula 4), recall (Formula 5), and F1 (Formula 6). Precision refers to the proportion of the actual PTs in the predicted PTs, recall is the proportion of the predicted PTs in the actual PTs, and F1 value is a combination of precision and recall.

$$P = \frac{tp}{tp + tn} \tag{4}$$

$$R = \frac{tp}{tp + fn} \tag{5}$$

$$F1 = \frac{2^*P^*R}{(P + R)} \tag{6}$$

where tp is the number of true positives, fp is the number of false positives, tn is the number of true negatives, and fn is the number of false negatives.

To select the most suitable algorithm for our dataset to ensure the effectiveness of our MLIFPR approach, we compare four traditional machine learning models that are widely used to complete the promising technology recognition task, including the random forest (RF), support vector machine (SVM), logistic regression (LR), and artificial neural network (ANN). We employ bibliometric, text, and IAHP features to train selected four machine learning methods, their performance of promising technology recognition is exhibited in Table A7 in Appendix 3. From Table A7, we can conclude that selecting RF to recognize promising technologies is effective compared with other algorithms.

Meanwhile, we provide the statistical analysis of collected bibliometric features and IAHP features for observing whether exist feature differences between promising and nonpromising technologies. The statistical description of bibliometric indicators and IAHP indicators are exhibited in Table A2 and Table A3 of Appendix 2, respectively. In extracting text features, the feature dimension of EC, Ship, RC, EP, and EV fields is set to 15, 8, 10, 15, and 8 according to the optimal topic number in each field (see Fig. A1 in Appendix 2). Furthermore, we employ Doc2Vec to obtain different dimensional text features used to recognize promising technologies. Then we observe that the text representation model does not provide a significant improvement to the performance of promising technology recognition (see Table A4 in Appendix 2), but leads to poorer model interpretability. Meanwhile, we employ the SHapley Additive exPlanations value to survey key text topics affecting promising technology recognition (see Fig. A2 in Appendix 2), then discover promising technologies favor on some special topics. And text description of each topic is exhibited in Table A5 of Appendix 2.

3.4. Evaluation

Our experiment is divided into three parts. First, we employ various feature combinations to train the MLIFPR approach for searching optimal feature combination and exploring IAHP features' effectiveness. These feature combinations are merged from bibliometric, text, and IAHP features of technologies. Next, we separately employ the inventor's and the assignee's historic performance features to train the MLIFPR approach for further analyzing IAHP features. Finally, we compare the MLIFPR approach with methods are proposed by previous studies for an effective assessment.

3.4.1. Performance under different feature combinations

To compare the performance of MLIFPR approaches that employ different feature combinations for training, judge the effectiveness of IAHP features, and observe multidimensional features whether possess optimal performance. We input different feature combinations of technologies to train and generate various MLIFPR approaches. These MLIFPR approaches are the same except for the input feature combinations.

Table 3 shows the promising technology recognition results of various MLIFPR approaches with different feature combinations in training. To all selected fields, we observe that the feature combination consisting of the bibliometric, text, and IAHP features shows optimal performance compared with other feature combinations. For example, in the RC field, the performance of the MLIFPR approach employing the IAHP, bibliometric, text, bibliometric and IAHP, text and IAHP, bibliometric and text, and bibliometric, text, and IAHP features are approximately 0.93, 0.91, 0.90, 0.95, 0.94, 0.94, and 0.96, respectively. These experimental results indicate that extracting multidimensional features to complete promising technology recognition is indispensable because of its optimal performance.

In addition, from Table 3, we discover that the performance of the bibliometric and IAHP features is superior to the bibliometric features,

Table 3
The performance comparison of different feature combinations.

Dataset	Metrics		Feature combination						
			bibliometric	IAHP	text	bibliometric+IAHP	text+IAHP	bibliometric+text	bibliometric+text+IAHP
EC	PT	P	0.8962	0.8874	0.8725	0.9523	0.9578	0.9639	0.9591
		R	0.8903	0.8492	0.8927	0.9474	0.9186	0.9254	0.9439
		F1	0.8932	0.8679	0.8825	0.9499	0.9378	0.9442	0.9515
	NPT	P	0.8910	0.8554	0.8902	0.9477	0.9218	0.9282	0.9448
		R	0.8969	0.8923	0.8696	0.9526	0.9595	0.9654	0.9597
		F1	0.8939	0.8735	0.8797	0.9501	0.9403	0.9464	0.9522
Ship	PT	P	0.8975	0.9127	0.9131	0.9404	0.9326	0.9482	0.9498
		R	0.9315	0.9437	0.8976	0.9837	0.9668	0.9559	0.9821
		F1	0.9142	0.9280	0.9053	0.9616	0.9494	0.9521	0.9657
	NPT	P	0.9288	0.9418	0.8993	0.9829	0.9655	0.9556	0.9815
		R	0.8936	0.9099	0.9146	0.9376	0.9302	0.9478	0.9481
		F1	0.9109	0.9256	0.9069	0.9598	0.9475	0.9517	0.9645
RC	PT	P	0.9145	0.9234	0.9087	0.9455	0.9239	0.9507	0.9455
		R	0.9229	0.9410	0.9020	0.9748	0.9727	0.9361	0.9768
		F1	0.9187	0.9322	0.9054	0.9599	0.9476	0.9433	0.9609
	NPT	P	0.9222	0.9399	0.9027	0.9740	0.9711	0.9371	0.9760
		R	0.9137	0.9220	0.9094	0.9438	0.9198	0.9515	0.9437
		F1	0.9179	0.9308	0.9060	0.9587	0.9448	0.9442	0.9596
EP	PT	P	0.9056	0.9037	0.9004	0.9493	0.9360	0.9658	0.9489
		R	0.9046	0.9278	0.8986	0.9712	0.9576	0.9253	0.9742
		F1	0.9051	0.9156	0.8995	0.9602	0.9467	0.9451	0.9614
	NPT	P	0.9047	0.9258	0.8988	0.9706	0.9566	0.9283	0.9735
		R	0.9057	0.9012	0.9006	0.9482	0.9345	0.9672	0.9475
		F1	0.9052	0.9133	0.8997	0.9592	0.9454	0.9473	0.9603
EV_uspto	PT	P	0.9545	0.9232	0.9002	0.9345	0.9315	0.9746	0.9369
		R	0.9066	0.9532	0.9033	0.9779	0.9745	0.9192	0.9775
		F1	0.9299	0.9380	0.9017	0.9557	0.9525	0.9461	0.9568
	NPT	P	0.9110	0.9516	0.9029	0.9769	0.9733	0.9236	0.9765
		R	0.9568	0.9207	0.8998	0.9314	0.9283	0.9760	0.9342
		F1	0.9333	0.9359	0.9014	0.9536	0.9503	0.9491	0.9549
EV_epo	PT	P	0.9619	0.9237	0.9002	0.9430	0.9306	0.9796	0.9388
		R	0.8964	0.9478	0.9047	0.9700	0.9756	0.9058	0.9756
		F1	0.9280	0.9356	0.9025	0.9563	0.9525	0.9413	0.9568
	NPT	P	0.9030	0.9464	0.9043	0.9691	0.9743	0.9125	0.9746
		R	0.9645	0.9217	0.8998	0.9414	0.9272	0.9811	0.9364
		F1	0.9327	0.9339	0.9020	0.9551	0.9502	0.9456	0.9551

and the increased performance exceeds 0.04. The performance of the text and IAHP features exceeds the text features, and the improved performance exceeds 0.04. The performance of bibliometric, text, and IAHP features is superior to the bibliometric and text features, and the outperformed performance exceeds 0.02. The performance of the IAHP features is the highest compared to the bibliometric or text features. For example, in the RC field, the P, R, and F1 of the bibliometric and IAHP features are approximately 0.95 and that of the bibliometric features are approximately 0.91. The P, R, and F1 of the text and IAHP features are approximately 0.94 and that of the text features are approximately 0.90. The P, R, and F1 of the bibliometric, text, and IAHP features are approximately 0.96 and that of the bibliometric and text features are approximately 0.94. The P, R, and F1 of the IAHP, bibliometric, and text features are approximately 0.93, 0.91, and 0.90, respectively. These comparison results prove the necessity of extracting the IAHP features considering their use promotes the performance improvement of promising technology recognition. The performance of the IAHP features outperforms the bibliometric and text features, which also indicates that extracting IAHP features is vital.

In summary, the combination of the bibliometric, text, and IAHP features is the optimal proxy of technologies' promisingness. Given that its optimal performance compared with other feature combinations. The introduction of IAHP features can improve the performance of various feature combinations. Thus, it is indispensable to employ multidimensional features and introduce the IAHP features to recognize promising technologies.

3.4.2. *Further analysis of IAHP features*

To further analyze the key factors in IAHP features, we employ the inventor's historic performance features, the assignee's historic performance features, and the IAHP features to train MLIFPR approaches to complete the promising technology recognition task. These MLIFPR approaches are the same except for the input feature combinations.

Table 4 shows the promising technology recognition results of different IAHP feature combinations. From Table 4, the performance of IAHP features outperforms the inventor's historic performance features or the assignee's historic performance features. For example, in the EC field, the P, R, and F1 of the inventor's historic performance, assignee's historic performance, and IAHP features are approximately 0.79, 0.66, and 0.86, respectively. These results show that employing the IAHP features is the optimal choice, and using only the inventor's or the assignee's historic performance features does not result in better performance.

In addition, we observe that the performance of the inventor's historic performance features is superior to the assignee in the EC, ship, RC, EP, and EV fields. It indicates that the inventor's historic performance features can better represent the promisingness of technologies (at least in these five fields). The possible explanation is that inventors are the source of technological innovation, and assignees are only the agents of technology. This result indicates that the weight of the inventor's historic performance indicators should be higher compared with the assignee when employing the IAHP indicators to construct a linear model to score the technology's promisingness. We should pay more attention to the inventor's R&D capability rather than the assignee when recognizing promising technologies. It is advisable to look for a star inventor rather than a star assignee when intending to develop promising technologies.

To detect the key predictive factors that most significantly affect promising technology recognition among IAHP features, we use the

Table 4
The comparison of different IAHP feature combinations.

Dataset	Metrics		Inventor's historic performance	Assignee's historic performance	IAHP
EC	PT	P	0.8375	0.7158	0.8874
		R	0.7516	0.6274	0.8492
		F1	0.7922	0.6687	0.8679
	NPT	P	0.7747	0.6684	0.8554
		R	0.8542	0.7509	0.8923
		F1	0.8125	0.7072	0.8735
Ship	PT	P	0.9157	0.8356	0.9127
		R	0.9431	0.8237	0.9437
		F1	0.9292	0.8296	0.9280
	NPT	P	0.9413	0.8262	0.9418
		R	0.9132	0.8380	0.9099
		F1	0.9270	0.8320	0.9256
RC	PT	P	0.9202	0.8486	0.9234
		R	0.9321	0.8553	0.9410
		F1	0.9261	0.8519	0.9322
	NPT	P	0.9312	0.8541	0.9399
		R	0.9192	0.8473	0.9220
		F1	0.9252	0.8507	0.9308
EP	PT	P	0.8983	0.7457	0.9037
		R	0.9281	0.8463	0.9278
		F1	0.9129	0.7928	0.9156
	NPT	P	0.9256	0.8223	0.9258
		R	0.8949	0.7114	0.9012
		F1	0.9100	0.7628	0.9133
EV_uspto	PT	P	0.9166	0.8247	0.9232
		R	0.9468	0.8165	0.9532
		F1	0.9315	0.8206	0.9380
	NPT	P	0.9450	0.8183	0.9516
		R	0.9139	0.8264	0.9207
		F1	0.9292	0.8223	0.9359
EV_epo	PT	P	0.9043	0.7650	0.9237
		R	0.9500	0.7917	0.9478
		F1	0.9266	0.7781	0.9356
	NPT	P	0.9474	0.7842	0.9464
		R	0.8995	0.7567	0.9217
		F1	0.9228	0.7702	0.9339

SHapley Additive exPlanations (SHAP) values to investigate the importance rank of all IAHP indicators. Fig. 4 shows the SHAP values of the IAHP indicators. It reveals that MCPI, HCPI, and SCPI are the most important indicators for promising technology recognition tasks. This result indicates that the forward citation information of inventors' technologies contributes greatly to promising technology recognition.

3.4.3. Comparison with baseline approaches

Performance comparison with previous studies needs to be implemented for an effective assessment of our proposed MLIFPR approach. Accordingly, we searched studies with similar purposes of recognizing promising technologies, and then introduced their labeling method, collected technology features, employed machine learning model, and performance. The datasets used in existing approaches are the same as our MLIFPR approach. Considering that most previous studies collect patents from USPTO or EPO to complete promising technology recognition, we select patents from the EV field in this study to complete the approach comparison. Table 5 shows the performance comparison result, it can be seen that the MLIFPR approach possesses optimal performance compared with other studies in promising technology recognition. The most persuasive explanation is our proposed approach adopted multidimensional features (bibliometric, text, and IAHP) as predictive factors distinguish from the previous studies focusing frequently on bibliometric and text features.

4. Discussion

Promising technologies can break the existing technological ecosystem, and then build a fresh industry structure. Hence, the purpose of this study is that propose an approach framework (MLIFPR approach) for recognizing promising technologies to assist R&D workers in capturing future technological reform. This study does not intend to develop a sophisticated machine learning algorithm with optimal promising technology recognition performance but to provide a method of extracting multidimensional features (e.g., IAHP). We believe that the

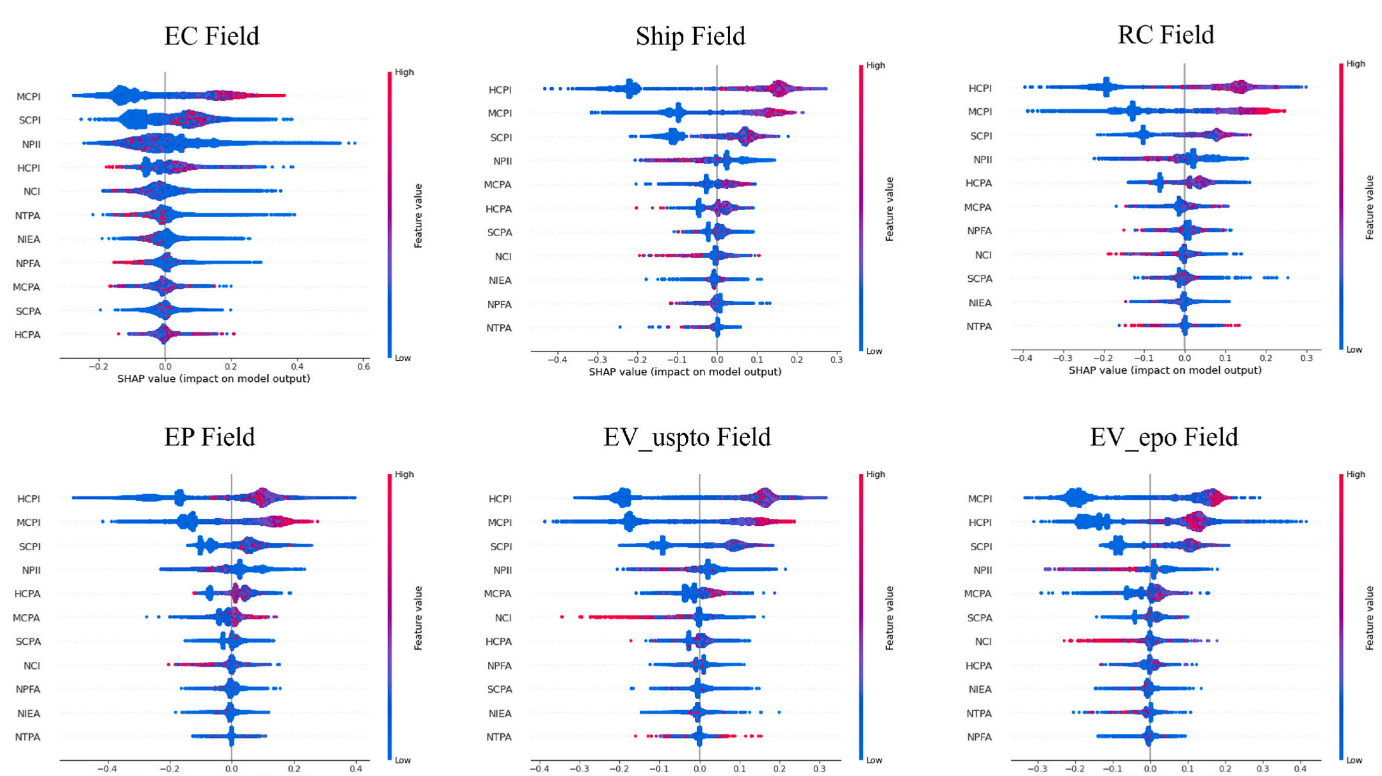

Fig. 4. The importance rank of IAHP indicators.

Table 5
Performance comparison with previous studies.

Dataset	Existing studies	Labeling method	Patent features	Employed model	Performance		
					P	R	F1
EV_uspto	Choi et al., 2021	forward citations	32 indicators	XGB	0.9676	0.9258	0.9462
	Zhou et al., 2021	forward citations	11 indicators	DNN	0.9699	0.9072	0.9375
	Kwon and Geum., 2020	forward citations	17 indicators	RF	0.9446	0.9576	0.9511
	MLIFPR approach	forward citations	bibliometric, text, and IAHP	RF	0.9369	0.9775	0.9568
EV_epo	Choi et al., 2021	forward citations	32 indicators	XGB	0.9672	0.9099	0.9377
	Zhou et al., 2021	forward citations	11 indicators	DNN	0.9474	0.9461	0.9467
	Kwon and Geum., 2020	forward citations	17 indicators	RF	0.9738	0.9041	0.9377
	MLIFPR approach	forward citations	bibliometric, text, and IAHP	RF	0.9388	0.9756	0.9568

MLIFPR approach can provide references for technological innovation works. Here, we list some potentially useful management implications of the MLIFPR approach.

4.1. Collecting multidimensional technology features

This study proves that extracting multidimensional features is vital for promising technology recognition tasks. Given that collecting more predictive factors can exploit the advantage of machine learning approaches in processing high-dimensional data. Hence, the collected features of technologies can be expanded when employing the MLIFPR approach. In addition to promising technology recognition, the MLIFPR approach can complete the automated patent classification (Li et al., 2018) to relieve the manual stress in patents' field division. The MLIFPR approach may provide references to the core patent identification task (Milanez et al., 2017) by employing multidimensional patent indicators.

Practical perspective, the MLIFPR approach may provide new insights for governments to capture promising technologies, and then assist them in developing emerging industries and achieving rapid economic growth (Altuntas et al., 2020; Oh et al., 2020; Zhang and Yu, 2020). The MLIFPR approach may be useful to enterprises that intend to develop promising technologies, due to its assistance in technological reform opportunities' capture. (Jang et al., 2021; Lee and Sohn, 2021; Zhu et al., 2019). Besides, recognizing promising technologies can speed up the technological R&D process of enterprises, and put investment and efforts into the place where science and technology breakthroughs are faster (Weis and Jacobson, 2021).

4.2. Employing IAHP features in patent analysis

Our experiments show that the IAHP features of technologies are indispensable in promising technology recognition tasks. Meanwhile, our proposed IAHP features can provide insights for other technological innovation works that intend to explore inventors' and assignees' R&D capabilities. For example, the IAHP features can be used for evaluating the transfer or litigation possibility of patents (Hu and Zhang, 2021; Chen et al., 2019; Kim et al., 2021). In the process of seeking competitors or partners (Qi et al., 2022), the IAHP features can assist enterprises know others' R&D strength in the current market. In studies related to auctioned patents (Lee and Lee, 2010), the IAHP features can find patents with high promisingness, then assist enterprises adjust patents' auction prices in time. When seeking star inventors or assignees (Zhu et al., 2019), the IAHP features can provide a tool to analyze their R&D capabilities. The IAHP features can provide a solution to analyze inventor team members' technological innovation competitiveness, and then expand the analysis perspectives of studies exploring inventor team formation rules (Jiao et al., 2022). When studying technological innovation ecosystems' stability, the IAHP features can provide a series of quantified indicators to measure their technological competitiveness to evaluate their risk resistance ability (Lancker et al., 2016).

Besides the patent analysis, the IAHP features proposed in this study can expand the feature extraction dimensions of other knowledge product analysis tasks, including scientific papers (Weis and Jacobson, 2021), open-source software (Zhao and Wei, 2017), and scientific projects (Zhu et al., 2022).

5. Conclusion

In this study, a novel approach framework to recognize promising technologies is proposed. Furthermore, the possibility of efficiently integrating multidimensional features and adding IAHP features in promising technology recognition is explored. To achieve this, patents in EC, Ship, RC, EP, and EV fields are collected and used to validate the effectiveness of this framework. According to experimental results, we obtain two main conclusions.

First, to technological innovation works, this study develops a quantitative model to capture the inventor and assignee's innovative potentials, namely the IAHP features of technologies. The experimental results show that IAHP features can improve the performance of promising technology recognition. Thus, it is an indispensable part of representing technologies' promisingness. Furthermore, we split IAHP features into inventors' and assignees' historic performance features, then discover that their employment alone does not lead to better performance of promising technology recognition. And the performance of inventors' historic performance features outperforms assignees in all datasets. In addition, we discover that the forward citation information of inventors' technologies contributes greatly to recognizing promising technologies through the SHAP values analysis. We think that the IAHP features can also provide a reference for other technological innovation works in expanding their feature selection dimensions.

Second, a machine learning framework named MLIFPR is proposed to recognize promising technologies. This framework provides a reference for fusing multidimensional features of technologies including bibliometric, text, and IAHP. Our experiments indicate that employing multidimensional features to recognize promising technologies is the optimal choice compared with using other feature combinations among bibliometric, text, and IAHP features. The MLIFPR framework can become a novel tool for practical activities related to promising technology recognition. For example, it can assist enterprises grasp technological reform opportunities and earning vast commercial profits by cultivating promising technologies in advance. With the use of this tool, governments can achieve rapid economic growth by developing emerging industries. To young inventors, this framework can point out cutting edge R&D directions for them, provide a chance to strengthen their R&D capabilities, and assist them in accelerating the completion of technological breakthroughs.

Despite the meaningful contributions of this study, it is more for a feasibility test of feature extension in recognizing promising technologies. Accordingly, the proposed approach has several limitations. First, considering that proposing multidimensional features is our main work, we only employ a few patent indicators to recognize promising technologies. The use of other patent indicators (e.g., cosine similarity between text vectors of patent claims, patent novelty) should be helpful for recognizing promising technologies. Second, the reviewers of this paper point out that an organization may have several people with the same names, or employees may change their affiliations. Hence, the inventor's name disambiguation method of using the inventor's and assignee's names needs to be improved in future work. In brief, we propose an approach framework of fusing multidimensional features to complete the promising technology recognition task. Simultaneously, the use of IAHP features provides a new reference for identifying other knowledge products' promisingness, including papers, open-source software, and research projects.

CRediT authorship contribution statement

Liang Gui: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Jie Wu:** Writing – review & editing, Resources, Investigation, Conceptualization. **Peng Liu:** Writing – review & editing, Methodology, Formal analysis. **Tieju Ma:** Writing – review & editing.

Ethics statements

No animal studies are presented in this manuscript.

No human studies are presented in this manuscript.

No potentially identifiable human images or data are presented in this study.

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Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Appendix 1. The acronym table

See Table A1.

Table A1
The acronym table.

Acronym	Full name
IPC	International Patent Classification
IAHP	Inventor and assignee's historical performance
MLIFPR	a machine-learning approach integrating bibliometric, text, and IAHP features for promising technology recognition
NWFC	The number of words of the first claim
NC	The number of claims
NA	The number of assignees
NI	The number of inventors
NIPC	The number of International Patent Classifications
NSFP	The number of simple family patents
NEFP	The number of extended family patents
NPP	The number of the priority patents
NBC	The number of backward citations
NCP	The number of citing papers
NPII	The number of patents invented by the inventor
SCPI	The sum of forward citations of patents of the inventor
MCPI	The mean of forward citations of patents of the inventor
HCPI	The highest forward citations of patents of the inventor
NCI	The number of coinventors
NPFA	The number of patents filed by the assignee
SCPA	The sum of forward citations of patents of the assignee
MCPA	The mean of forward citations of patents of the assignee
HCPA	The highest of forward citations of patents of the assignee
NTPA	The number of transferred patents of the assignee
NIEA	The number of inventors entrusted to the assignees
LDA	latent Dirichlet allocation
PTs	promising technologies
NPTs	nonpromising technologies
CNIPA	China National Intellectual Property Administration
EC	Electrical communication
RC	Road construction
EP	Electric power
EV	Electric vehicle
SVM	support vector model
RF	random forest
LR	logistic regression
ANN	artificial neural networks
R&D	research and development

Appendix 2. Extraction of bibliometric, IAHP, and text features

See Table A2, Table A3, Table A4, Table A5, Fig. A1, and Fig. A2.

Table A2
Statistical description of technologies' bibliometric features.

Dataset	bibliometric features	Sample		PTs Subsample		NPTs Subsample		t-test
		mean	standard deviation	mean	standard deviation	mean	standard deviation	
EC	NWFC	415.74	438.53	270.55	295.02	431.87	448.78	−59.30***
	NC	10.08	8.33	10.58	6.61	10.02	8.50	10.62***
	NA	1.09	0.36	1.10	0.37	1.09	0.36	3.31***
	NI	2.86	2.22	2.83	2.23	2.86	2.22	−2.71***
	NIPC	2.34	3.58	2.32	1.08	2.34	3.76	−1.04**
	NSFP	2.55	3.17	1.52	1.68	2.67	3.28	−58.66***
	NEFP	2.86	7.34	1.59	2.77	3.00	7.66	−30.73***
	NPP	0.05	0.30	0.02	0.18	0.06	0.31	−19.63***
	NBC	4.29	2.05	4.54	2.28	4.26	2.02	21.74***
	NCP	0.36	0.92	0.40	0.99	0.36	0.92	7.47***
Ship	NWFC	403.77	421.10	308.71	338.71	414.33	427.98	−6.83***
	NC	5.29	3.51	5.90	3.59	5.23	3.50	5.27***
	NA	1.11	0.44	1.10	0.37	1.11	0.45	−0.25
	NI	3.54	2.87	3.79	2.85	3.51	2.87	2.57***
	NIPC	1.86	1.25	2.04	1.51	1.84	1.21	4.41***
	NSFP	1.81	2.19	1.16	0.68	1.88	2.28	−8.98***
	NEFP	1.85	2.34	1.18	0.85	1.92	2.44	−8.64***
	NPP	0.02	0.15	0.01	0.17	0.02	0.15	−1.18
	NBC	5.94	2.35	6.24	2.41	5.91	2.34	3.78***
	NCP	0.39	1.08	0.67	1.50	0.36	1.01	7.71***
RC	NWFC	393.88	417.57	296.93	397.24	404.65	418.39	−10.44***
	NC	5.50	3.76	6.24	3.98	5.41	3.73	8.88***
	NA	1.16	0.55	1.19	0.63	1.16	0.54	2.00*
	NI	4.27	3.59	4.49	3.41	4.24	3.61	2.79***
	NIPC	2.08	1.65	2.44	2.08	2.04	1.59	9.96***
	NSFP	1.81	1.62	1.24	0.75	1.87	1.67	−15.94***
	NEFP	1.98	3.46	1.28	1.45	2.05	3.61	−9.03***
	NPP	0.02	0.22	0.02	0.20	0.03	0.22	−1.46
	NBC	5.69	2.27	5.50	2.42	5.71	2.26	−3.73***
	NCP	0.42	1.11	0.77	1.49	0.38	1.05	14.18***
EP	NWFC	488.27	533.43	357.92	401.54	502.75	544.19	−25.84***
	NC	5.96	4.97	6.60	4.36	5.89	5.02	13.71***
	NA	1.27	0.68	1.28	0.68	1.27	0.68	0.56
	NI	3.75	3.16	3.99	3.00	3.72	3.18	7.86***
	NIPC	2.03	1.30	2.17	1.34	2.01	1.29	11.49***
	NSFP	1.97	3.11	1.46	5.37	2.03	2.73	−17.45***
	NEFP	2.22	7.22	1.62	7.95	2.29	7.13	−8.78***
	NPP	0.04	0.23	0.03	0.18	0.04	0.24	−5.76***
	NBC	4.61	2.18	4.51	2.23	4.62	2.17	−4.85***
	NCP	0.40	0.96	0.63	1.22	0.37	0.93	25.11***
EV_uspto	NWFC	1091.57	847.00	1086.02	1172.67	1092.19	802.70	−0.42
	NC	20.53	111.23	43.48	294.84	17.98	63.43	13.32***
	NA	1.39	1.07	1.39	1.16	1.39	1.06	−0.27
	NI	2.56	1.84	2.84	2.12	2.53	1.80	10.05***
	NIPC	4.27	4.11	5.58	5.59	4.12	3.88	20.64***
	NSFP	4.64	5.27	5.42	11.12	4.55	4.13	9.58***
	NEFP	7.97	31.64	15.34	49.33	7.15	28.90	15.04***
	NPP	1.22	1.16	1.48	1.95	1.19	1.03	14.47***
	NBC	17.06	29.28	26.44	54.80	16.01	24.66	20.75***
	NCP	1.19	7.63	2.42	15.80	1.05	6.06	10.45***
EV_epo	NWFC	997.28	819.18	922.44	866.02	1005.60	813.39	−4.31***
	NC	13.51	19.10	12.32	14.03	13.64	19.58	−2.94***
	NA	1.08	0.34	1.07	0.31	1.08	0.35	−0.73
	NI	2.54	1.81	2.54	1.81	2.54	1.81	0.05
	NIPC	4.70	3.88	4.76	4.03	4.69	3.87	0.83
	NSFP	5.98	4.50	5.29	5.55	6.06	4.36	−7.28
	NEFP	7.63	16.28	6.73	21.60	7.73	15.58	−2.61**
	NPP	1.25	0.96	1.22	0.93	1.25	0.97	−1.15
	NBC	4.37	3.95	4.35	4.14	4.38	3.93	−0.24
	NCP	0.38	1.22	0.31	1.68	0.39	1.15	−2.55*

*** means $p < 0.01$.
** means $p < 0.05$.
* means $p < 0.1$.

Table A3
Statistical description of technologies' IAHP features.

Dataset	IAHP features	Sample		PTs Subsample		NPTs Subsample		t-test
		mean	standard deviation	mean	standard deviation	mean	standard deviation	
EC	NPII	18.87	43.55	16.00	42.98	19.19	43.60	−11.76***
	SCPI	6.97	24.98	12.32	34.07	6.37	23.68	38.26***
	MCPI	0.45	1.02	1.60	2.31	0.32	0.63	216.62***
	HCPI	2.49	4.27	4.64	5.28	2.25	4.08	91.15***
	NCI	11.55	19.57	9.97	19.29	11.73	19.59	−14.43***
	NPFA	6511.77	11,480.65	3903.84	8933.27	6801.53	11,693.73	−40.56***
	SCPA	1503.07	2767.17	1165.41	2546.63	1540.59	2788.08	−21.74***
	MCPA	0.63	0.78	1.05	1.49	0.58	0.64	97.13***
	HCPA	15.52	13.98	15.56	13.55	15.52	14.02	0.52
	NTPA	865.10	1541.59	520.89	1231.92	903.35	1567.57	−39.86***
	NIEA	5779.13	9838.11	3483.53	7622.60	6034.19	10,021.77	−41.67***
Ship	NPII	4.79	7.86	3.91	4.92	4.89	8.11	−3.40***
	SCPI	8.08	14.71	19.48	16.12	6.82	13.99	24.20***
	MCPI	2.07	3.69	8.44	6.92	1.36	2.20	63.66***
	HCPI	4.98	8.03	13.56	8.70	4.03	7.36	34.48***
	NCI	6.05	9.43	6.58	9.79	5.99	9.39	1.69
	NPFA	31.91	55.71	30.13	51.69	32.11	56.13	−0.97
	SCPA	59.16	103.15	75.08	101.67	57.39	103.16	4.66***
	MCPA	2.07	2.81	5.83	5.49	1.65	1.91	45.23***
	HCPA	11.46	13.73	18.10	12.36	10.73	13.68	14.78***
	NTPA	2.60	6.21	1.92	4.62	2.68	6.36	−3.34***
	NIEA	72.14	146.32	73.89	143.83	71.95	146.60	0.36
RC	NPII	5.26	7.26	4.50	6.31	5.35	7.35	−4.73***
	SCPI	9.33	17.32	22.81	22.55	7.83	15.95	36.14***
	MCPI	2.16	3.71	8.85	6.89	1.41	2.13	101.10***
	HCPI	5.16	7.57	14.09	8.15	4.17	6.82	57.52***
	NCI	8.45	13.72	8.69	12.70	8.42	13.83	0.80
	NPFA	26.59	49.57	26.16	45.30	26.63	50.02	−0.39
	SCPA	51.51	95.19	72.37	102.20	49.19	94.09	9.85***
	MCPA	2.14	2.89	6.20	5.87	1.69	1.85	71.31***
	HCPA	10.80	12.42	17.99	11.22	10.00	12.29	26.47***
	NTPA	4.20	16.83	2.64	11.34	4.37	17.32	−4.16***
	NIEA	58.31	113.96	63.55	113.95	57.73	113.94	2.06***
EP	NPII	10.76	28.39	9.29	18.99	10.92	29.24	−5.46***
	SCPI	20.25	53.73	40.39	73.15	18.01	50.63	39.83***
	MCPI	2.07	3.73	7.86	7.69	1.43	2.19	190.91***
	HCPI	6.83	10.77	16.48	13.03	5.76	9.93	98.93***
	NCI	9.66	17.74	10.87	17.61	9.52	17.75	7.24***
	NPFA	640.61	2076.51	661.51	2080.15	638.28	2076.10	1.06
	SCPA	1153.02	3470.82	1259.18	3478.38	1141.22	3469.78	3.22***
	MCPA	2.05	2.42	4.68	5.20	1.76	1.62	123.43***
	HCPA	24.84	30.48	32.84	29.26	23.95	30.48	27.76***
	NTPA	60.68	184.46	61.98	183.66	60.54	184.55	0.74
	NIEA	1282.87	4477.62	1322.69	4484.50	1278.45	4476.83	0.94
EV_uspto	NPII	3.83	9.71	4.79	11.62	3.73	9.47	6.32***
	SCPI	82.73	285.25	236.59	477.55	65.64	249.29	35.31***
	MCPI	22.84	41.67	88.44	96.74	15.55	18.94	119.06***
	HCPI	36.62	72.14	125.64	144.74	26.73	49.77	87.17***
	NCI	5.61	17.96	11.93	38.44	4.91	13.76	22.80***
	NPFA	204.30	408.63	174.56	375.11	207.61	412.05	−4.69***
	SCPA	3104.36	5964.29	3136.84	5660.66	3100.75	5997.06	0.35
	MCPA	22.88	36.62	69.82	91.84	17.67	16.77	91.27***
	HCPA	126.05	163.05	206.98	204.31	117.06	155.22	32.40***
	NTPA	16.35	42.11	18.55	45.20	16.10	41.74	3.36***
	NIEA	250.91	442.61	215.29	407.54	254.87	446.16	−5.18***
EV_epo	NPII	3.12	7.61	3.75	10.68	3.05	7.18	3.91***
	SCPI	5.22	24.19	17.14	36.96	3.89	21.93	23.56***
	MCPI	1.52	4.20	7.85	9.28	0.82	2.26	82.19***
	HCPI	2.92	6.91	11.11	11.24	2.01	5.54	60.87***
	NCI	3.20	4.75	3.30	4.91	3.19	4.73	0.98
	NPFA	181.25	357.02	175.75	365.84	181.86	356.02	−0.73
	SCPA	234.38	496.44	264.85	535.17	230.99	491.83	2.90***
	MCPA	1.52	2.76	4.11	6.51	1.23	1.70	46.67***
	HCPA	19.96	23.78	25.39	23.83	19.35	23.70	10.81***
	NTPA	7.17	14.04	6.39	12.60	7.26	14.18	−2.63***
	NIEA	276.12	499.41	260.18	492.23	277.89	500.17	−1.51

*** means $p < 0.01$.
** means $p < 0.05$.
* means $p < 0.1$.

Table A4
Performance under different dimensional text features.

Dataset	Metrics		Dimension				
			8	16	32	64	128
EC	PT	P	0.8730	0.8747	0.8758	0.8836	0.8990
		R	0.8923	0.8963	0.8846	0.8843	0.8788
		F1	0.8826	0.8854	0.8802	0.8840	0.8888
	NPT	P	0.8908	0.8932	0.8828	0.8838	0.8821
		R	0.8713	0.8710	0.8739	0.8832	0.9019
		F1	0.8810	0.8820	0.8783	0.8835	0.8919
Ship	PT	P	0.8882	0.9078	0.9228	0.9343	0.9203
		R	0.9083	0.9239	0.8871	0.9216	0.9171
		F1	0.8982	0.9158	0.9046	0.9279	0.9187
	NPT	P	0.9080	0.9230	0.8883	0.9235	0.9191
		R	0.8879	0.9068	0.9237	0.9360	0.9222
		F1	0.8979	0.9148	0.9057	0.9297	0.9207
RC	PT	P	0.9045	0.9080	0.9027	0.9140	0.9229
		R	0.8985	0.9049	0.9094	0.8971	0.9009
		F1	0.9015	0.9064	0.9060	0.9055	0.9118
	NPT	P	0.8984	0.9063	0.9095	0.8997	0.9001
		R	0.9045	0.9094	0.9028	0.9162	0.9223
		F1	0.9014	0.9078	0.9061	0.9079	0.9111
EP	PT	P	0.8993	0.8942	0.8866	0.8991	0.9146
		R	0.8970	0.8938	0.8913	0.8853	0.8830
		F1	0.8981	0.8940	0.8889	0.8921	0.8985
	NPT	P	0.8969	0.8946	0.8921	0.8860	0.8870
		R	0.8992	0.8950	0.8875	0.8998	0.9176
		F1	0.8980	0.8948	0.8898	0.8929	0.9020
EV_uspto	PT	P	0.9002	0.9036	0.9125	0.9150	0.9323
		R	0.9033	0.9107	0.9080	0.8950	0.9038
		F1	0.9017	0.9071	0.9102	0.9049	0.9178
	NPT	P	0.9029	0.9099	0.9090	0.8981	0.9057
		R	0.8998	0.9027	0.9134	0.9175	0.9337
		F1	0.9014	0.9062	0.9112	0.9077	0.9195
EV_epo	PT	P	0.9002	0.8879	0.8993	0.9146	0.9111
		R	0.9047	0.9015	0.8961	0.8942	0.9078
		F1	0.9025	0.8947	0.8977	0.9043	0.9095
	NPT	P	0.9043	0.9010	0.8933	0.8923	0.9099
		R	0.8998	0.8873	0.8966	0.9130	0.9131
		F1	0.9020	0.8941	0.8950	0.9025	0.9115

Table A5
Description of text topics in each field.

Field	Topic	Description
EC	Topic0	Communication of terminal devices.
	Topic1	The operation of the call function and short messaging service function.
	Topic2	The transmission and processing of video data and image data.
	Topic3	Matrix processing algorithm based on coded information.
	Topic4	The storage unit and reading unit of data.
	Topic5	The reception unit and processing unit of radio signals.
	Topic6	Short-distance radio information transmission based on routers and relay stations.
	Topic7	Digital signal demodulation and acceptance in different frequencies and channels.
	Topic8	The control system and management module of wireless communication.
	Topic9	Chip and processor system of radio signal transmission base station.
	Topic10	Encrypted transmission of radio signals between clients.
	Topic11	Design and operation of mobile camera hardware and software.
	Topic12	Transmission and conversion of photoelectric signal in optical fiber equipment.
	Topic13	Transmission protocol and terminal address of digital information.
	Topic14	Frequency modulation and conversion of digital signals by radio signal transmission base stations.
Ship	Topic0	The mooring device of ships.
	Topic1	The power system of ships.
	Topic2	The design of ships' keel and internal structure.
	Topic3	The connecting structure of ships.
	Topic4	Waterproofing system and emergency handling device of ships.
	Topic5	The bearing unit of ships.
	Topic6	The driving equipment of ships.
	Topic7	The design of deck units of ships.
RC	Topic0	The design and construction of bridge structure.
	Topic1	The protection system and warning equipment for urban main roads.
	Topic2	The design and construction of urban main roads.
	Topic3	The damping unit and protection system for bridges.
	Topic4	Materials used in the construction of urban main roads.

(*continued on next page*)

Table A5 (*continued*)

Field	Topic	Description
	Topic5	The design of cleaning equipment that is used for urban main roads.
	Topic6	The design and construction of roads for rail transit.
	Topic7	The sensing equipment and control system for rail transit roads.
	Topic8	Construction materials and connecting devices for rail transit roads.
	Topic9	The design of cleaning systems for urban roads, tunnels, overpasses, and other roads.
EP	Topic0	Solar power generation equipment.
	Topic1	The control device and conversion device of the electric transmission line.
	Topic2	The electric storage equipment and charge and discharge management of the power station.
	Topic3	The design of transformers.
	Topic4	Materials and connecting devices for electric transmission lines.
	Topic5	Power transmission management based on control and sensing devices.
	Topic6	Bearings and electromagnetic conversion devices of turbine generators.
	Topic7	Power grid dispatching management of balancing power generation output and power load.
	Topic8	Driving devices of power generation equipment.
	Topic9	The electromagnetic induction device of the generator.
	Topic10	High voltage electric transmission equipment.
	Topic11	Installing and connecting devices of electric transmission equipment.
	Topic12	Inverter equipment for electric transmission stations.
	Topic13	Protective devices for the electric transmission system.
	Topic14	Current and voltage conversion systems for electric transmission stations.
EV	Topic0	The R&D activities related to the electrode layer material in batteries of electric vehicles.
	Topic1	The brake system of the engine.
	Topic2	The drive shaft of the motor in electric vehicles.
	Topic3	The electric current storage control system of electric vehicles.
	Topic4	The connection of multiple control systems in electric vehicles.
	Topic5	The cooling unit and heating unit of electric vehicles.
	Topic6	The speed-controlling device of electric vehicles.
	Topic7	The signal transmission device of electric vehicles.

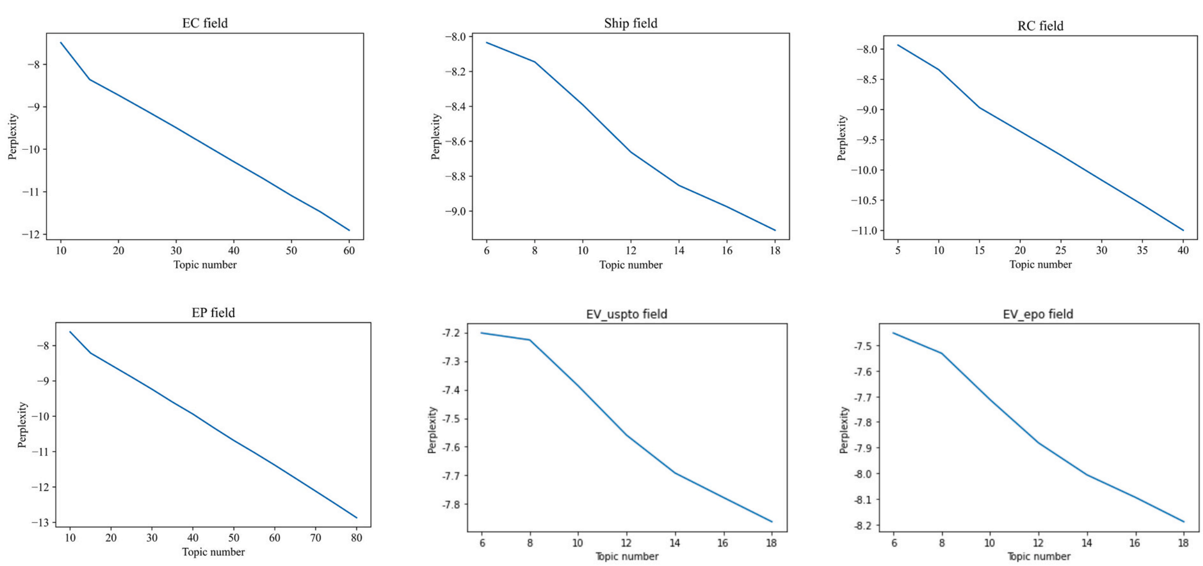

Fig. A1. The perplexity change curve under different topic number.

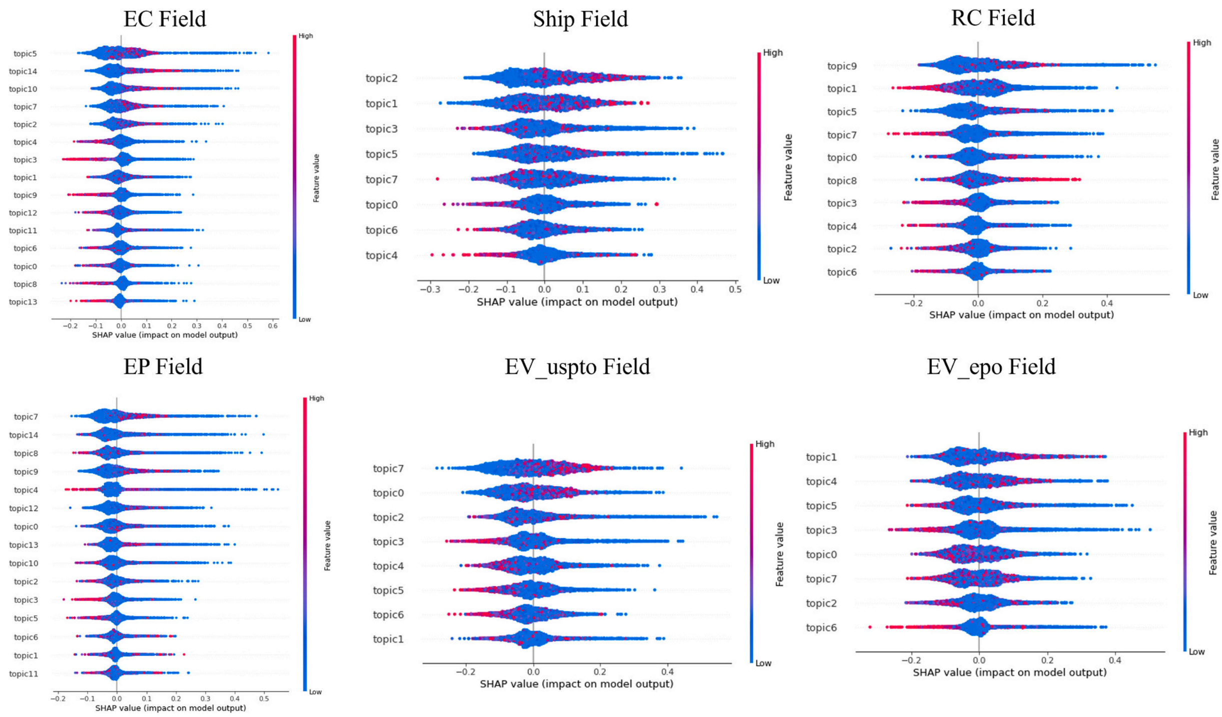

Fig. A2. The importance rank of text topics analyzed by SHAP values.

Appendix 3. Recognition of promising technologies

See Table A6, Table A7.

Table A6
Confusion Matrix.

		Predicted	
		Positive	Negative
Actual	Positive	True positive (TP)	False negative (FN)
	Negative	False-positive (FP)	True negative (TN)

Table A7
The comparison of different machine learning models.

Dataset	Metrics		Model			
			SVM	ANN	LR	RF
EC	PT	P	0.7860	0.7836	0.7933	0.9591
		R	0.8307	0.8083	0.7847	0.9439
		F1	0.8077	0.7957	0.7890	0.9515
	NPT	P	0.8180	0.7994	0.7843	0.9448
		R	0.7709	0.7739	0.7929	0.9597
		F1	0.7938	0.7864	0.7885	0.9522
Ship	PT	P	0.8985	0.9014	0.8817	0.9498
		R	0.9687	0.9694	0.8716	0.9821
		F1	0.9323	0.9342	0.8766	0.9657
	NPT	P	0.9663	0.9671	0.8738	0.9815
		R	0.8912	0.8946	0.8838	0.9481
		F1	0.9272	0.9295	0.8788	0.9645
RC	PT	P	0.8916	0.8861	0.8638	0.9455
		R	0.9718	0.9586	0.8743	0.9768
		F1	0.9300	0.9209	0.8690	0.9609
	NPT	P	0.9703	0.9569	0.8780	0.9760
		R	0.8866	0.8818	0.8677	0.9437
		F1	0.9266	0.9178	0.8728	0.9596
EP	PT	P	0.8565	0.8394	0.8711	0.9489
		R	0.9325	0.9313	0.8373	0.9742

(*continued on next page*)

Table A7 (*continued*)

Dataset	Metrics		Model			
			SVM	ANN	LR	RF
		F1	0.8929	0.8830	0.8538	0.9614
	NPT	P	0.9245	0.9214	0.8407	0.9735
		R	0.8410	0.8187	0.8740	0.9475
		F1	0.8808	0.8670	0.8570	0.9603
EV_uspto	PT	P	0.8989	0.8645	0.9011	0.9369
		R	0.9415	0.9530	0.8385	0.9775
		F1	0.9197	0.9066	0.8687	0.9568
	NPT	P	0.9379	0.9470	0.8475	0.9765
		R	0.8929	0.8491	0.9070	0.9342
		F1	0.9148	0.8954	0.8762	0.9549
EV_epo	PT	P	0.9005	0.8673	0.8978	0.9388
		R	0.9381	0.9577	0.8075	0.9756
		F1	0.9189	0.9102	0.8503	0.9568
	NPT	P	0.9366	0.9537	0.8282	0.9746
		R	0.8983	0.8563	0.9098	0.9364
		F1	0.9171	0.9024	0.8671	0.9551

Data availability

Data will be made available on request.

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